Assignment 19.2 Ai Assisted Coding

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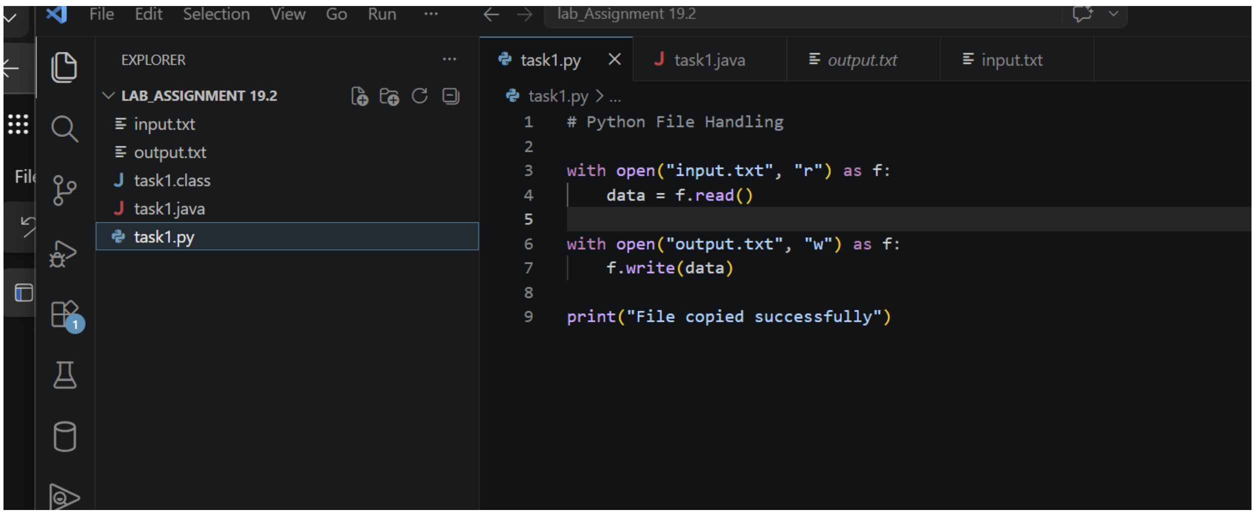
Btno:06

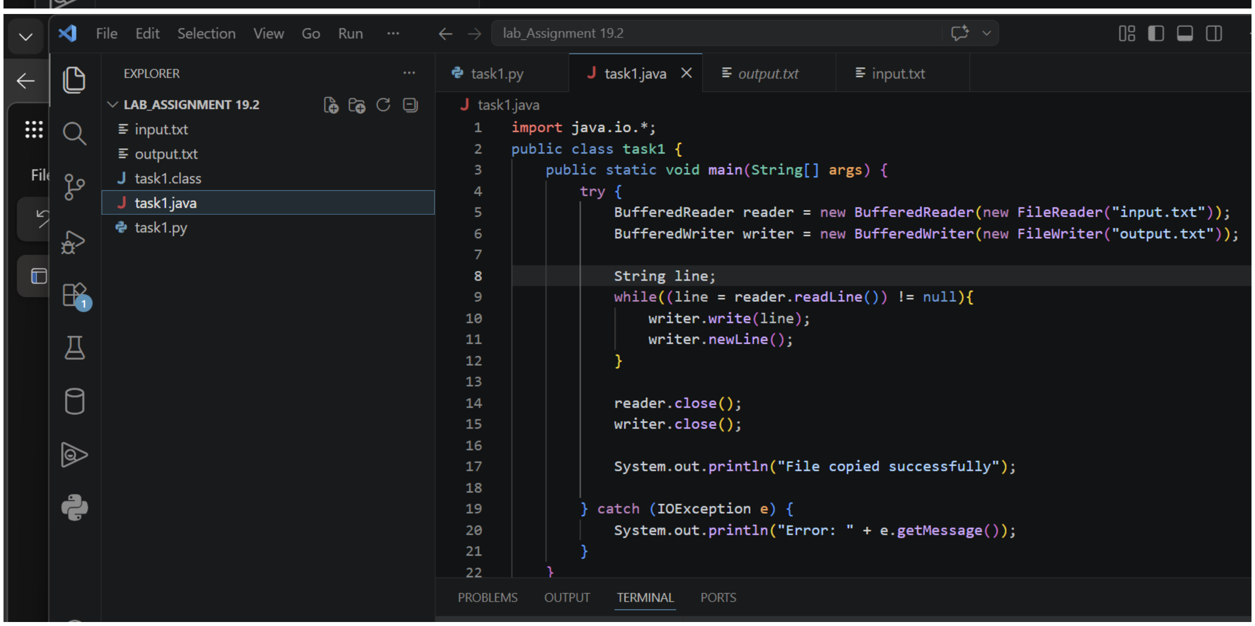
Task 1: Python to Java Conversion (File Handling)

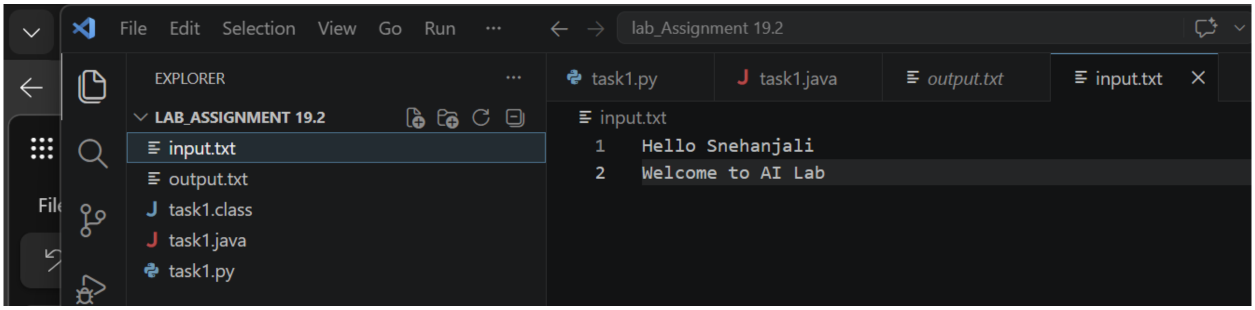
Prompt:

Convert a Python file handling program into Java.

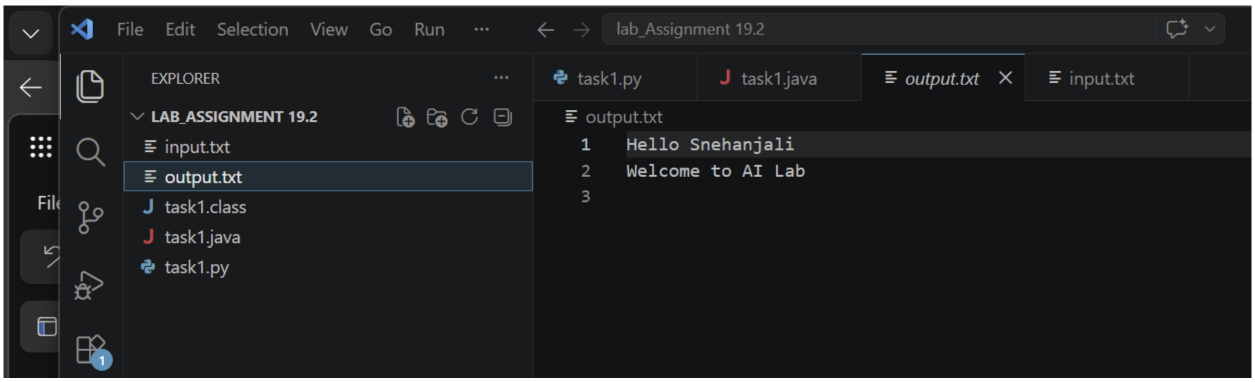
Code:

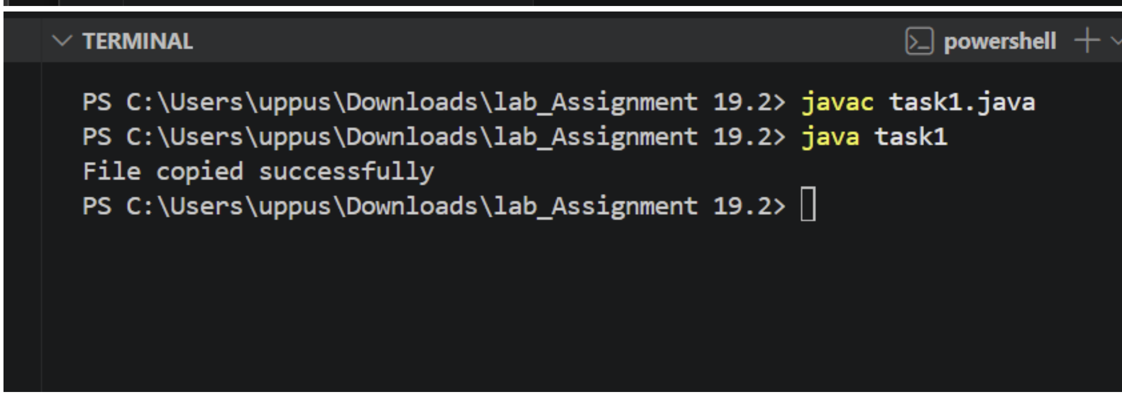






Output:





Explanation:

The Python program reads data from a file and writes it to another file using open() function.

The Java program uses BufferedReader and BufferedWriter for file handling.

Exception handling is required in Java.

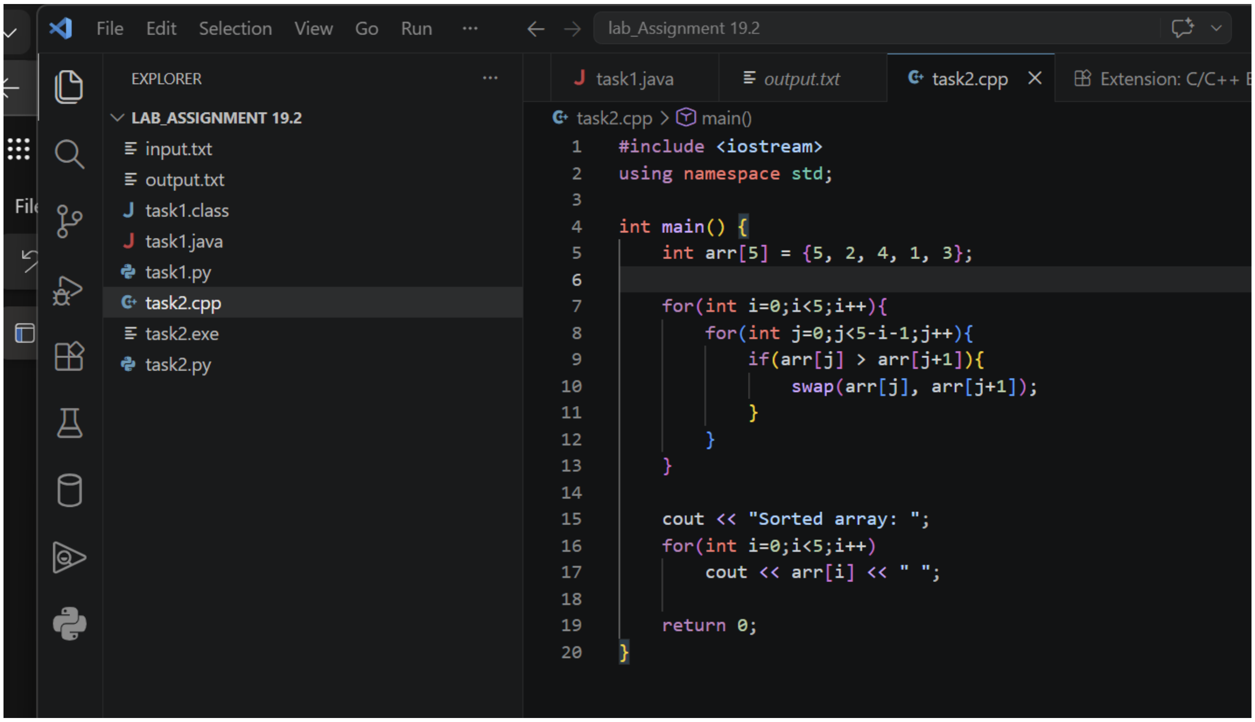
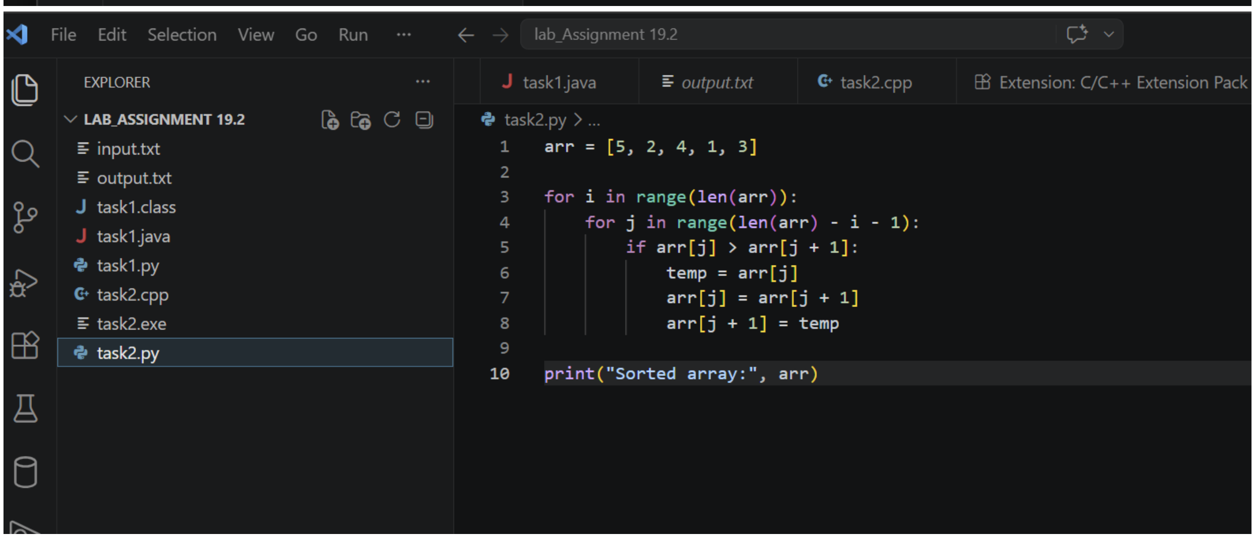
Both programs perform the same operation and produce identical output.

Task2: C++ to Python Conversion (Sorting)

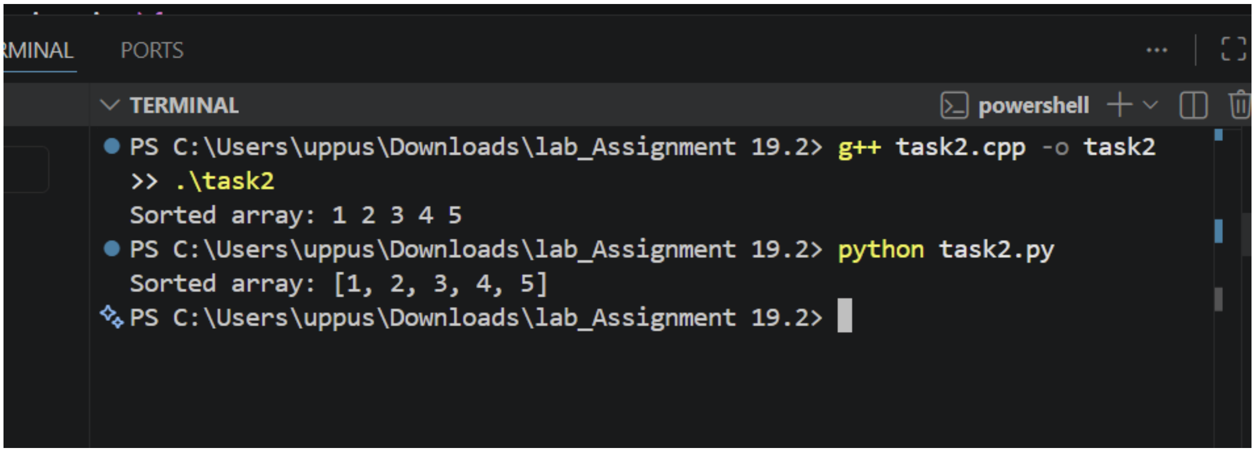
Prompt:

Convert a C++ sorting program into Python.

Code:


File Edit Selection View Go Run ···
lab_Assignment 19.2
EXPLORER
LAB_ASSIGNMENT 19.2
input.txt
output.txt
task1.class
task1.java
task1.py
task2.cpp
task2.exe
task2.py
task1.java
output.txt
task2.cpp
Extension: C/C++ E
task2.cpp > main()
#include <iostream>
using namespace std;
int main() {
int arr[5] = {5, 2, 4, 1, 3};
for(int i=0;i<5;i++){
for(int j=0;j<5-i-1;j++){
if(arr[j] > arr[j+1]){
swap(arr[j], arr[j+1]);
}
}
}
cout << "Sorted array: ";
for(int i=0;i<5;i++)
cout << arr[i] << " ";
return 0;
}

File Edit Selection View Go Run ···
lab_Assignment 19.2
EXPLORER
LAB_ASSIGNMENT 19.2
input.txt
output.txt
task1.class
task1.java
task1.py
task2.cpp
task2.exe
task2.py
task1.java
output.txt
task2.cpp
Extension: C/C++ Extension Pack
task2.py > ...
arr = [5, 2, 4, 1, 3]
for i in range(len(arr)):
for j in range(len(arr) - i - 1):
if arr[j] > arr[j + 1]:
temp = arr[j]
arr[j] = arr[j + 1]
arr[j + 1] = temp
print("Sorted array:", arr)

Output:



Explanation:

-->The C++ program uses Bubble Sort to sort an array.

-->The Python program follows the same logic.

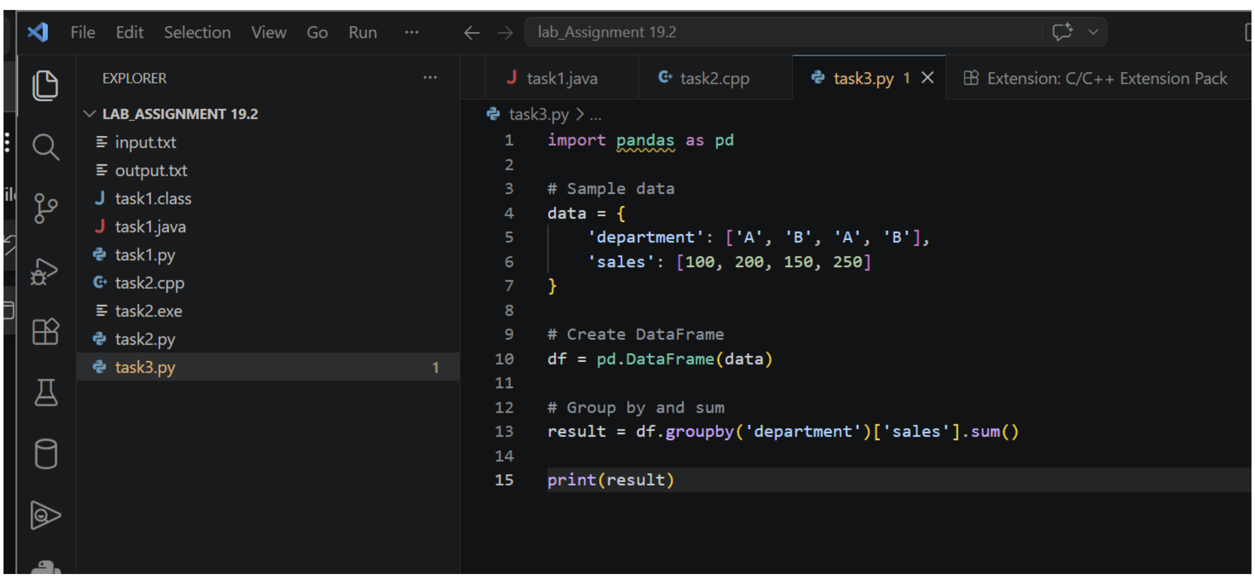
-->Both programs produce the same sorted result.

Task 3: SQL to Pandas Conversion

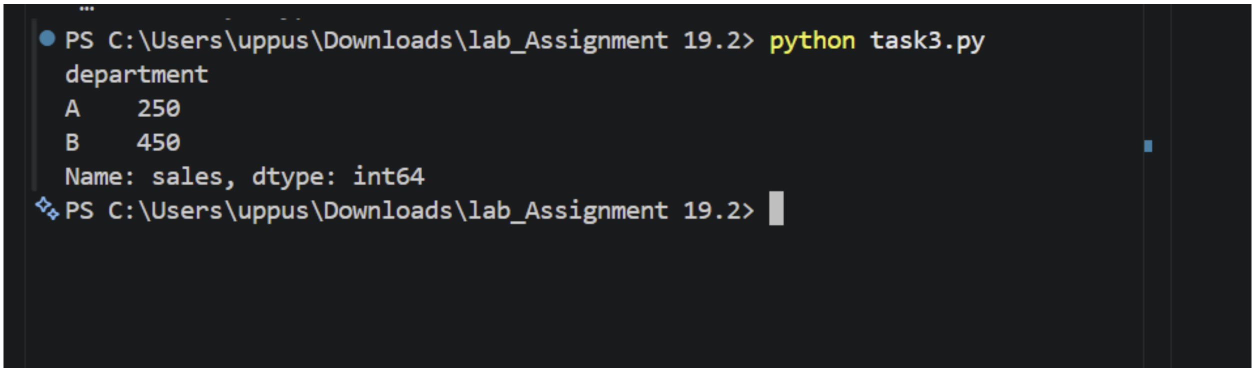
Prompt:

Convert a SQL query with GROUP BY into Pandas.

Code:



Output:



Explanation:

-->The SQL query groups data by department and calculates total sales.

-->The Pandas code uses groupby() and sum() to achieve the same result.

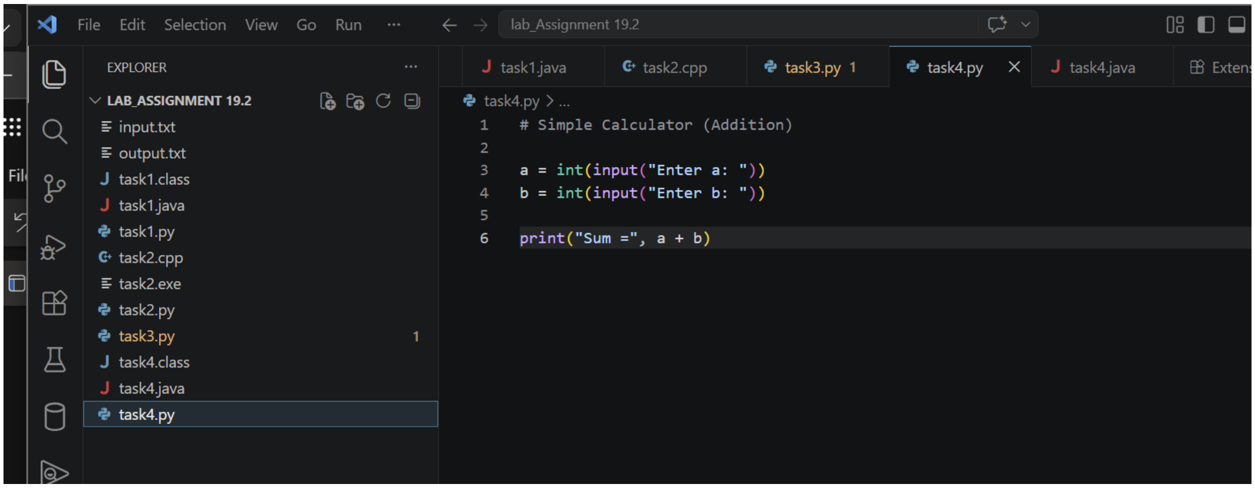
-->Both produce identical outputs.

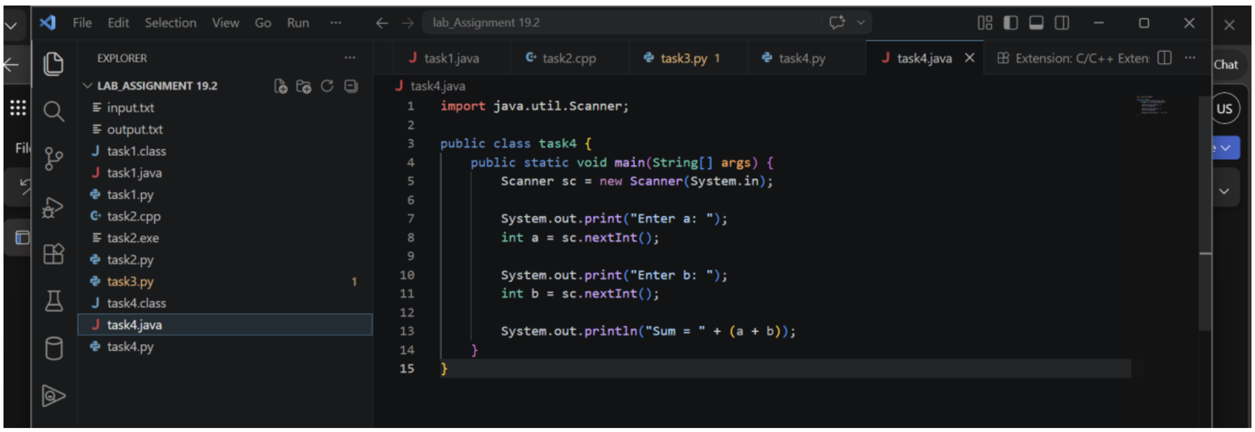
Task 4: Real-Time Code Conversion (Python to Java)

Prompt:

Convert a Python calculator program into Java.

Code:





Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task4.py
Enter a: 3
Enter b: 5
Sum = 8
```

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> javac task4.java
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> java task4
Enter a: 3
Enter b: 5
Sum = 8
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The Python program performs addition using simple input and print statements.

-->The Java program uses Scanner for input and follows class-based structure.

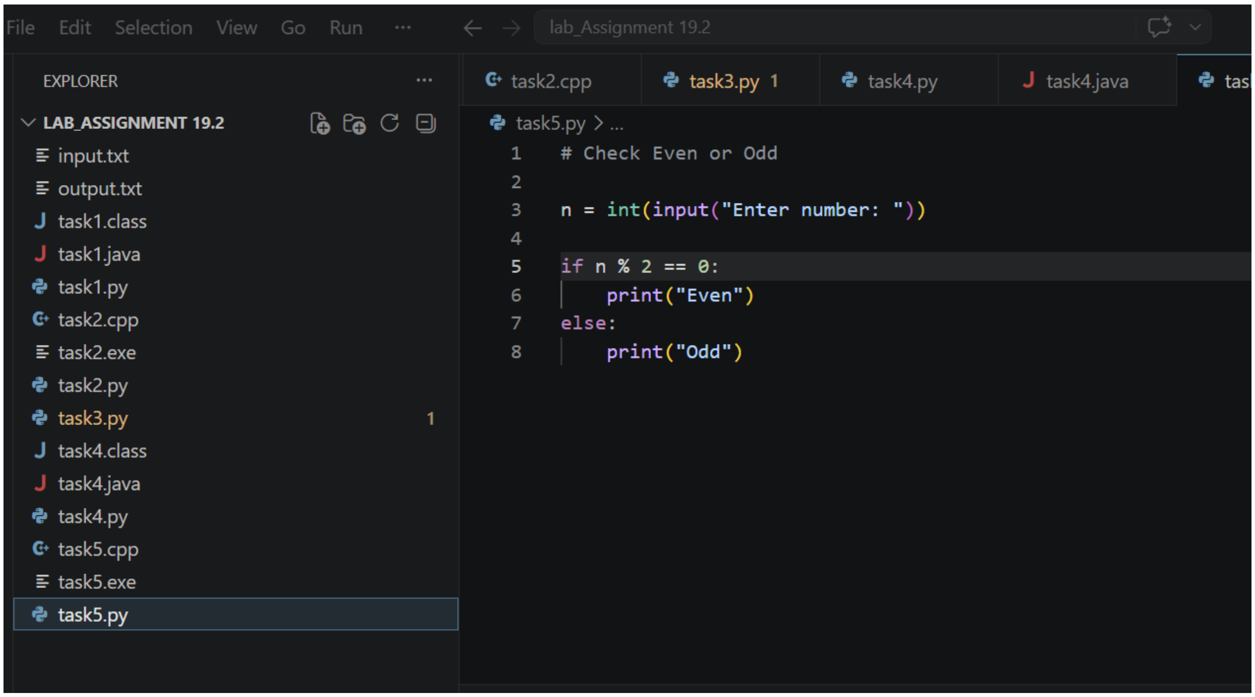
-->Both programs perform the same operation and produce identical output.

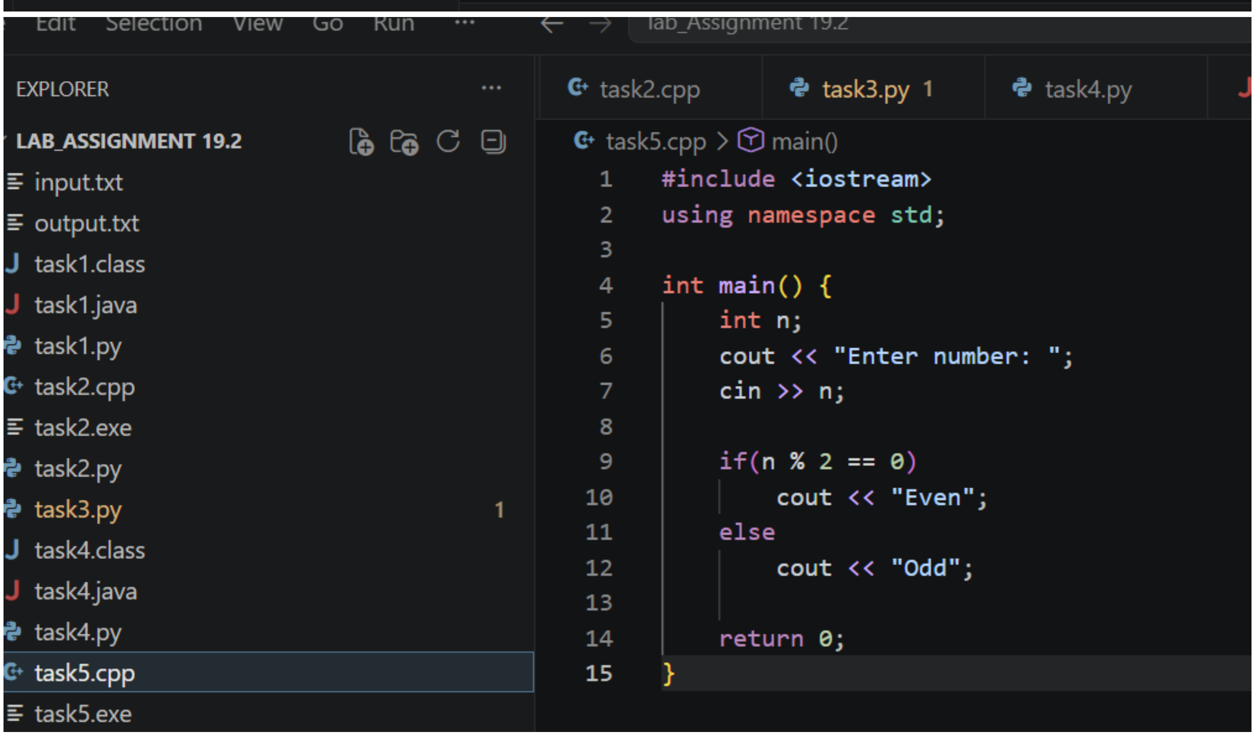
Task5: Python to C++ Conversion (Even/Odd)

Prompt:

Convert a Python program to C++ to check even or odd number.

Code:





Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task5.py
Enter number: 3
Odd
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task5.cpp -o task5
>> .\task5
Enter number: 3
Odd
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The Python program checks whether a number is even or odd using modulus operator.

-->The C++ program follows the same logic using if-else condition.

-->Both programs produce correct results for all test cases.